

PROJECT 1 — DATA CLEANING & PREPARATION

PDF Change Log · Blessing Dan · DecodeLabs Batch 2026

Intern	Blessing Dan
Organization	DecodeLabs
Batch	2026
Project	Project 1 — Data Cleaning & Preparation
Dataset	E-Commerce Orders — 1,200 Records 14 Columns
Date	May 2026
Status	Complete ✓

PROJECT OVERVIEW

Project 1 is the foundation phase of the DecodeLabs Data Analytics Internship. The goal is to clean a raw dataset by handling missing values, removing duplicates, and correcting data formats. As DecodeLabs states: *"Before you build a single predictive model or dashboard, you must master the art of scrubbing raw, messy datasets into reliable sources of truth."* This change log documents every transformation applied — what was found, what was done, and why. **If it isn't documented, it didn't happen.**

DECODELABS VERIFICATION GATE

Before completing Project 1, DecodeLabs requires proof of zero duplicate IDs and zero incorrectly formatted dates — a 0% error rate on both checks.

Verification Check	Requirement	Result	Status
Unique Identifiers (OrderID)	0% Error Rate	Zero duplicates found	✓ PASSED
Date Formats (YYYY-MM-DD)	0% Error Rate	Zero format errors found	✓ PASSED

CHANGE LOG

Every change made to the dataset during cleaning and preparation is documented below in the official DecodeLabs format.

Change ID	Description	Impact	Status
CR001	Loaded dataset into Python using <code>pd.read_csv()</code>	Dataset loaded successfully 1,200 rows × 14 columns confirmed	✓ Resolved
CR002	Checked all 14 columns for missing/null values using <code>isnull().sum()</code>	CouponCode: 309 missing (25.8%) All other 13 columns: 0 missing	✓ Resolved
CR003	Filled CouponCode missing values with "No Coupon" using <code>fillna("No Coupon")</code>	309 missing values replaced with "No Coupon" — no data lost 0 missing values remain	✓ Resolved
CR004	Verified CouponCode column after filling missing values	CouponCode missing values: 0 All 1,200 records now complete	✓ Resolved
CR005	Checked OrderID column for duplicate records using <code>duplicated().sum()</code>	Zero duplicates found 0% Error Rate ✓ Threshold met	✓ Resolved
CR006	Converted Date column from string to datetime format using <code>pd.to_datetime()</code>	Date column now in YYYY-MM-DD format — time analysis enabled	✓ Resolved
CR007	Verified date format consistency across all 1,200 date entries	Zero format errors found 0% Error Rate ✓ Threshold met	✓ Resolved
CR008	Created Month column from Date using <code>dt.to_period('M')</code>	Monthly revenue trend analysis enabled for Project 2 EDA	✓ Resolved
CR009	Created Year column from Date using <code>dt.year</code>	Yearly comparison enabled across 2023, 2024 and 2025	✓ Resolved
CR010	Verified numeric columns: Quantity, UnitPrice, TotalPrice	No negative values found All numeric data confirmed valid	✓ Resolved
CR011	Verified all data types for all columns using <code>df.dtypes</code>	All columns have correct types Dataset ready for Project 2 EDA	✓ Resolved

CLEANING SUMMARY

Metric	Result
Total Records	1,200
Total Columns	14
Duplicate OrderIDs	0 — None Found

Date Format Errors	0 — None Found
Missing Values Found	CouponCode: 309 (25.8%)
Missing Values Action	Filled with "No Coupon" — 0 missing values remain
Columns Converted	Date → datetime Month & Year created
Final Missing Values	0 across all 14 columns ✓
Final Dataset Status	Clean & Ready for Project 2 EDA ✓

Conclusion: The dataset has been successfully cleaned and prepared. All 309 missing values in CouponCode were filled with "No Coupon" — preserving all 1,200 records with zero data loss. Zero duplicate OrderIDs and zero date format errors confirm the dataset meets the DecodeLabs verification threshold (0% Error Rate on both checks). The dataset is now a reliable source of truth and is fully ready for Project 2 — Exploratory Data Analysis.